

ESGBuddy — ESG Frameworks & Ground Truth Reference

Automated Multi-Framework ESG Compliance Verification
Capstone Project — MPSTME, NMIMS University, 2026

Covers: BRSR · GRI · TCFD · SASB

Sections: Framework Details · Compliance Methodology · Ground Truth Evidence

PART I — ESG Reporting Frameworks

ESGBuddy evaluates company sustainability reports against four ESG reporting frameworks. Each framework has a distinct issuing body, mandatory vs. voluntary status, target audience, compliance philosophy, and clause structure. Understanding these differences is essential to understanding why ESGBuddy cannot use a single evaluation strategy for all four.

Framework	Issuing Body	Status	Compliance Philosophy
BRSR	SEBI (India)	Mandatory (NSE Top 1000)	Disclosure presence
GRI	GRI Foundation (Netherlands)	Voluntary (globally dominant)	Substantive coverage
TCFD	FSB / TCFD (International)	Voluntary → increasingly mandatory	Qualitative scenario analysis
SASB	IFRS Foundation (International)	Voluntary, industry-specific	Quantitative sector metrics

1. BRSR — Business Responsibility and Sustainability Reporting

1.1 What Is BRSR?

BRSR is India's mandatory ESG disclosure standard, introduced by the Securities and Exchange Board of India (SEBI) in 2021 and made compulsory for the top 1,000 NSE/BSE listed companies by market capitalisation from FY 2022–23. It replaced the older Business Responsibility Report (BRR) and significantly expanded the scope and granularity of required disclosures. The framework is deeply rooted in India's National Guidelines on Responsible Business Conduct (NGRBC), which sets out 9 Principles covering environmental sustainability, ethical governance, worker rights, community engagement, and consumer welfare.

1.2 Structure and Scope

BRSR is divided into three sections:

Section A — General Disclosures: Company identity, operations, products/services, number of employees (permanent and contractual), turnover, CSR spending, transparency and e-governance. These are largely boilerplate but mandatory.

Section B — Management and Process Disclosures: Policy statements, governance structures, and process descriptions for each of the 9 NGRBC Principles. Covers environmental policy, anti-corruption, stakeholder engagement, and supply chain disclosures.

Section C — Principle-wise Performance Disclosures: Quantitative and qualitative disclosures mapped to all 9 principles. This is the most substantive section. Subdivided into Essential (mandatory) and Leadership (voluntary but recommended) indicators.

The BRSR Core (introduced in FY 2023–24) defines a sub-set of ~50 Key Performance Indicators (KPIs) that are mandatory for the top 150 companies and will progressively expand. These include GHG footprint, water footprint, waste footprint, energy intensity, diversity ratios, and supply chain disclosures.

1.3 Key Disclosure Areas

Environmental: GHG emissions (Scope 1, 2, 3), energy consumption and intensity, water withdrawal and consumption, waste generated and disposed, biodiversity impact, environmental compliance violations

Social: Employee headcount (permanent/contract/male/female), turnover rate, maternity/paternity leave, average training hours, occupational health & safety incidents, POSH complaints

Governance: Board composition, anti-corruption policy, whistleblower mechanism, CSR spend, related-party transactions, regulatory penalties

Supply Chain: Percentage of suppliers assessed for environmental and social criteria, supply chain grievances

1.4 How ESGBuddy Checks BRSR Compliance

BRSR compliance philosophy: DISCLOSURE PRESENCE. SEBI cares that the company has disclosed the required information — not whether the disclosed values are good or sufficient. A company that discloses '0 tonnes of Scope 1 GHG' or 'Not applicable — no manufacturing operations' satisfies the clause. ESGBuddy's BRSR-specific prompt reflects this:

- Supported = clear, direct disclosure present — data, narrative, table, cross-reference with page number, 'Nil' with reason, or 'Not applicable' with reason.
- Partial = disclosure is present but incomplete, only implied by broader policy, or a key element of a multi-part indicator is missing.
- Not Supported = no disclosure, no proxy, field left blank.
- The rule engine checks: numeric values present (for quantitative KPIs), year references (for temporal reporting), required keywords (e.g. 'Scope 1', 'intensity'), and labeled field patterns (e.g. 'Total water consumed:').

ESGBuddy evaluates 140+ BRSR clauses per company. For 3 Indian companies (RIL, TATA Motors, TCS), the team created full-coverage ground truth across all 306 parsed BRSR clauses.

1.5 Why BRSR Scores Highest (F1 87.6%)

BRSR's structured, checkbox-style format means evidence is either present or absent in a clearly labeled section. Indian companies that file BRSR compliantly tend to use standardised table formats that are easy to retrieve and match. The disclosure-presence philosophy also makes the verdict boundary cleaner — partial cases are those with genuinely incomplete multi-part indicators.

2. GRI — Global Reporting Initiative

2.1 What Is GRI?

GRI is the world's most widely used voluntary sustainability reporting standard, developed by the Global Reporting Initiative — an independent international organisation founded in Boston in 1997, now headquartered in Amsterdam. As of 2024, over 10,000 organisations in 100+ countries use GRI. The GRI Standards were significantly updated in 2021 (GRI Universal Standards) and progressively through 2024-25 (updated sector and topic standards).

2.2 Structure and Scope

GRI Standards are organised in three tiers:

Universal Standards (GRI 1, 2, 3): GRI 1: Foundation — principles for reporting (materiality, stakeholder inclusiveness, sustainability context, completeness). GRI 2: General Disclosures — organisation profile, governance, strategy, policies, stakeholder engagement, reporting practice. GRI 3: Material Topics — how to determine, manage, and report on material topics.

Sector Standards: Industry-specific standards covering Oil & Gas, Coal, Agriculture/Aquaculture/Fishing, Mining, and others. Sectors are required to apply the relevant sector standard for their industry. Includes GRI 11 (Oil & Gas), GRI 12 (Coal), GRI 13 (Agriculture).

Topic Standards: 200 Series (Economic): GRI 201 Economic Performance, 202 Market Presence, 203 Indirect Economic Impacts, 204 Procurement, 205 Anti-Corruption, 206 Anti-Competitive Behaviour, 207 Tax. 300 Series (Environmental): GRI 301 Materials, 302 Energy, 303 Water & Effluents, 304 Biodiversity, 305 Emissions, 306 Waste, 308 Supplier Environmental Assessment. 400 Series (Social): GRI 401-418 covering Employment, Labour Relations, OHS, Training, Diversity, Non-discrimination, Freedom of Association, Child Labour, Forced Labour, Security Practices, Indigenous Peoples, Communities, Supplier Social Assessment, Public Policy, Customer Health & Safety, Marketing & Labeling, Customer Privacy.

2.3 Materiality in GRI

GRI's central concept is materiality — companies are not required to report on every topic, only those deemed material (significant impact on economy, environment, or society, OR significant influence on stakeholder decisions). Companies define their own material topics through a stakeholder engagement process. This makes GRI more flexible but also harder to evaluate uniformly — a clause that is material for one company may be non-material for another.

2.4 How ESGBuddy Checks GRI Compliance

GRI compliance philosophy: SUBSTANTIVE COVERAGE. Key distinctions:

- Supported = evidence substantively addresses the clause — specific data, narrative, policy, table, or a cross-reference that clearly points to the required content.
- Partial = evidence is related but indirect, incomplete, or only partially answers the clause. Covers implied compliance, proxy metrics, broader strategy text without the specific disclosure, or >50% addressed but a key element missing.
- Not Supported = blank, no proxy, or explicit denial.

- 'Zero', 'Nil', 'Not applicable' with reason = Supported (GRI allows companies to explain non-applicability).
- Materiality principle: if a topic is disclosed as non-material with justification, ESGBuddy does not penalise absence.

2.5 Why GRI Scores 81.9% F1

GRI's flexibility (materiality-based reporting) makes evaluation harder. Companies report in varying levels of detail and some clauses are genuinely ambiguous — a disclosure that is substantive enough for one auditor may be partial for another. The Partial category is large for GRI (122 of 390 labels = 31%), reflecting the genuine ambiguity in this framework.

3. TCFD — Task Force on Climate-related Financial Disclosures

3.1 What Is TCFD?

TCFD was established in 2015 by the Financial Stability Board (FSB), an international body that monitors and makes recommendations about the global financial system, at the request of the G20. The framework was designed to help companies disclose climate-related risks and opportunities in a consistent, comparable way that is useful to investors, lenders, and insurance underwriters. Final recommendations were published in June 2017. As of 2024, TCFD reporting is mandatory or expected in the UK, EU (through CSRD), Japan, Hong Kong, Singapore, and New Zealand, and voluntary but widely adopted in the US.

Note: TCFD has been formally disbanded (October 2023), with monitoring responsibilities transferred to the IFRS Foundation. However, its four-pillar framework remains the de facto structure for climate risk disclosure globally, embedded in IFRS S2, CSRD, and other standards.

3.2 Structure — Four Pillars and 11 Recommended Disclosures

TCFD is organised around four thematic pillars, each with recommended disclosures:

Governance

a) Describe the board's oversight of climate-related risks and opportunities. b) Describe management's role in assessing and managing climate-related risks and opportunities.

Strategy

a) Describe the climate-related risks and opportunities the organisation has identified over the short, medium, and long term. b) Describe the impact of climate-related risks and opportunities on the organisation's businesses, strategy, and financial planning. c) Describe the resilience of the organisation's strategy, taking into consideration different climate-related scenarios, including a 2°C or lower scenario.

Risk Management

a) Describe the organisation's processes for identifying and assessing climate-related risks. b) Describe the organisation's processes for managing climate-related risks. c) Describe how processes for identifying, assessing, and managing climate-related risks are integrated into the organisation's overall risk management.

Metrics & Targets

a) Disclose the metrics used to assess climate-related risks and opportunities in line with its strategy and risk management process. b) Disclose Scope 1, Scope 2, and if appropriate, Scope 3 greenhouse gas (GHG) emissions, and the related risks. c) Describe the targets used to manage climate-related risks and opportunities and performance against targets.

3.3 Scenario Analysis — The Most Distinctive TCFD Requirement

TCFD's Strategy (c) recommendation — resilience under different climate scenarios including a 2°C or lower scenario — is the most distinctive and challenging requirement. Companies are expected to model how their business strategy holds up under physical risks (rising sea levels, extreme weather) and transition risks (carbon taxes, stranded assets, policy changes) across multiple time horizons (short: 0-3 years, medium: 3-10 years, long: 10+ years). Very few companies do this

rigorously — most provide qualitative descriptions of risk categories without quantified scenario analysis.

3.4 How ESGBuddy Checks TCFD Compliance

TCFD compliance philosophy: QUALITATIVE SCENARIO-BASED ASSESSMENT.

- Supported = the specific TCFD requirement is fully and substantively met in the retrieved excerpts — not just a generic climate mention.
- Partial = mentioned but incomplete, boilerplate ('we take climate risk seriously'), ambiguous, or missing key elements of THIS specific requirement.
- Not Supported = no relevant evidence in the excerpts; the requirement is clearly absent.
- Materiality note: when clauses touch metrics, Scope 3, targets, or ERM integration, the LLM must note whether the company defines material climate issues.
- The TCFD system prompt instructs the LLM to judge each clause as written (extracted from TCFD PDFs) — not to collapse all disclosures into 'one of eleven boxes'.
- ESGBuddy evaluates each TCFD clause individually, including sub-disclosures and guidance text extracted from the official TCFD PDFs.

3.5 Why TCFD Scores Lowest (F1 77.6%)

Three reasons: (1) Forward-looking qualitative statements are inherently ambiguous — the boundary between 'partial' and 'not supported' for scenario analysis is genuinely unclear. (2) Climate risk information is often spread across multiple sections of a long report and the 512-token chunking may not retrieve it all. (3) Companies frequently write vague, boilerplate climate language that looks like it addresses TCFD but does not meet the specificity the framework requires — the LLM sometimes accepts this as Partial when human annotators judged it Not Supported.

4. SASB — Sustainability Accounting Standards Board

4.1 What Is SASB?

SASB was founded in 2011 in San Francisco to develop industry-specific sustainability accounting standards for use in US securities filings. In 2022, SASB merged with the International Integrated Reporting Council (IIRC) to form the Value Reporting Foundation, which was then consolidated into the IFRS Foundation — making SASB standards now maintained under the same umbrella as IFRS S1 and IFRS S2 (which draws heavily from TCFD).

SASB's defining feature is industry specificity: it publishes 77 industry standards across 11 sectors, each defining the specific sustainability topics and metrics most likely to be financially material for companies in that industry. A software company and a coal mining company have completely different SASB metrics.

4.2 The 11 Sectors and Industries Covered

Sector	Key Industries
Consumer Goods	Apparel, Appliance Manufacturing, Building Products, E-Commerce, Household Products, Retailers, Toys
Extractives & Minerals	Coal, Construction Materials, Iron & Steel, Metals & Mining, Oil & Gas (E&P, Midstream, Refining, Services)
Financials	Asset Management, Commercial Banks, Consumer Finance, Insurance, Investment Banking, Mortgage Finance
Food & Beverage	Agricultural Products, Alcoholic Beverages, Food Retailers, Meat/Poultry/Dairy, Non-Alcoholic Beverages, Processed Foods, Restaurants
Health Care	Biotech & Pharma, Health Care Delivery, Health Care Distributors, Managed Care, Medical Equipment, Drug Retailers
Infrastructure	Electric Utilities & Power Generators, Gas Utilities, Home Builders, Real Estate, Waste Management, Water Utilities
Renewable Resources	Biofuels, Forestry, Fuel Cells, Meat/Poultry, Pulp & Paper, Solar, Wind
Resource Transformation	Aerospace & Defence, Chemicals, Containers & Packaging, Electrical Equipment, Industrial Machinery
Services	Advertising & Marketing, Casinos & Gaming, Education, Hotels & Lodging, Leisure Facilities, Media, Professional Services, Restaurants, Telecommunication Services
Technology & Communications	Electronic Manufacturing, Hardware, Internet Media & Services, Semiconductors, Software & IT Services, Telecommunication Services
Transportation	Air Freight & Logistics, Airlines, Auto Parts, Automobiles, Car Rental, Cruise Lines, Marine,

4.3 How SASB Metrics Work

Each SASB industry standard defines: (1) Disclosure Topics — the sustainability issues that are financially material for that industry (e.g. Data Security for software, GHG Emissions for airlines). (2) Accounting Metrics — specific quantitative or qualitative indicators for each topic, with units and definitions. (3) Activity Metrics — denominators for normalising performance data (e.g. revenue, number of employees, fleet size).

Example — Software & IT Services industry SASB metrics:

- TC-SI-130a.1: (1) Total energy consumed (MWh), (2) percentage grid electricity (%), (3) percentage renewable (%)
- TC-SI-130a.2: Discussion of long-term and short-term strategy to manage Scope 1 and Scope 2 GHG emissions
- TC-SI-220a.1: Description of approach to identifying and addressing data security risks
- TC-SI-330a.1: Percentage of employees that are (1) foreign nationals and (2) located offshore
- TC-SI-550a.1: Discussion of integration of environmental, social, and governance factors in investment management

4.4 How ESGBuddy Checks SASB Compliance

SASB compliance philosophy: QUANTITATIVE INDUSTRY-SPECIFIC DISCLOSURE. ESGBuddy is intentionally lenient because:

- Large companies report across many pages; retrieved chunks are narrow excerpts — absence of a metric in 5 chunks does not mean it's absent from the report.
- Supported = clear, substantive response to the topic — narrative, policy, process, table, OR quantitative data. Perfection not required.
- Partial = topic appears but is thin, mostly boilerplate, or missing an obvious core element.
- Not Supported = ONLY if no chunk has any plausible thematic link to the clause topic.
- Hard rule: if any chunk discusses the same underlying theme (energy, GHG, water, safety, data security, etc.), not_supported is forbidden — minimum Partial.
- Industry context: the LLM infers the company's sector and accepts equivalent disclosures under different wording (e.g. a tech company reporting 'renewable energy certificates' satisfies an energy metric even if SASB code is not cited).

4.5 Why SASB Scores 84.4% F1

Large tech companies (Apple, Amazon, Infosys) — which form 3 of our 13 SASB-evaluated companies — produce detailed, well-structured sustainability reports that align well with SASB metrics. The lenient prompt also means fewer false negatives. However, the SASB ground truth shows high Partial rates ($227/390 = 58\%$) because companies often address topics directionally without meeting every SASB sub-metric — consistent with what human annotators observed.

5. Side-by-Side Framework Comparison

Attribute	BRSR	GRI	TCFD	SASB
Issuing body	SEBI, India	GRI Foundation	FSB / IFRS Foundation	IFRS Foundation
Geographic focus	India (NSE/BSE listed)	Global (voluntary)	Global (increasingly mandatory)	Global (industry-specific)
Mandatory?	Yes — top 1,000 companies	No	Partly (UK, EU, Japan, NZ)	No
Primary audience	SEBI regulators, investors	All stakeholders	Investors, lenders, insurers	Investors, analysts
Update cycle	SEBI circulars (annual)	Ongoing (2021 overhaul)	Superseded by IFRS S2	Post-merger with IIRC
Clauses in ESGBuddy	140+	120+	40+	77
Evaluation philosophy	Disclosure presence	Substantive coverage	Qualitative scenario analysis	Quantitative sector metrics
Verdict boundary	Presence vs. absence	Substantive vs. indirect	Specific vs. generic/boilerplate	On-topic vs. off-topic
Partial rate (our GT)	10.2% of 1,218 labels	31.3% of 390 labels	43.1% of 390 labels	58.2% of 390 labels
Compliant rate (our GT)	73.0% of 1,218 labels	29.2% of 390 labels	34.9% of 390 labels	28.0% of 390 labels
System F1	87.6%	81.9%	77.6%	84.4%
Hardest clauses	Multi-part quantitative KPIs	Material topic coverage	Scenario resilience (2°C)	Industry-specific sub-metrics

PART II — Manual Labelling: Evidence and Justification

This section provides detailed evidence that the ground truth in ESGBuddy was manually created by the project team through careful reading of actual company sustainability reports — not generated by any AI system. It is intended to be used in a viva to demonstrate the rigour and legitimacy of the benchmark.

6.1 Total Labels Created

Framework	Label Count
BRSR	1218 labels across 13 companies
GRI	390 labels across 13 companies
SASB	390 labels across 13 companies
TCFD	390 labels across 13 companies
GRAND TOTAL	2388 manually created labels
Used in standardised evaluation	1,560 (top-30 per company per framework × 13 companies × 4 frameworks)

Total labels created: 2388. Of these, 1,560 are used in the standardised evaluation (top-30 highest-priority clauses per company per framework). The additional BRSR labels (306 per company for RIL, TATA Motors, TCS) represent full-coverage ground truth for three Indian benchmark companies covering all parsed BRSR clauses.

6.2 Per-Company, Per-Framework Label Breakdown

The following table shows the exact count of Compliant (C), Partial (P), and Non-Compliant (NC) labels for every company in every framework. Variation across companies proves human judgment — an AI would produce more uniform distributions.

BRSR

Company	Total	Compliant	Partial	Non-Compliant
Amazon	30	3	7	20
Apple	30	0	8	22
GPM	30	7	8	15
Givaudan	30	2	6	22
Himadri	30	5	7	18
Infosys	30	4	7	19
NYK	30	1	4	25
Nestle	30	0	6	24
RIL	306	285	20	1
Sasken	30	10	5	15
TATA Motors	306	285	21	0
TCS	306	284	16	6
Unilever	30	3	9	18
TOTAL	1218	889	124	205

GRI

Company	Total	Compliant	Partial	Non-Compliant
Amazon	30	9	12	9
Apple	30	9	8	13
GPM	30	11	8	11
Givaudan	30	6	12	12
Himadri	30	11	6	13
Infosys	30	8	12	10
NYK	30	3	10	17
Nestle	30	7	5	18
RIL	30	13	9	8
Sasken	30	11	8	11
TATA Motors	30	7	11	12
TCS	30	12	8	10
Unilever	30	7	13	10
TOTAL	390	114	122	154

SASB

Company	Total	Compliant	Partial	Non-Compliant
Amazon	30	0	29	1
Apple	30	22	8	0
GPM	30	0	22	8
Givaudan	30	0	26	4
Himadri	30	12	18	0
Infosys	30	1	26	3
NYK	30	30	0	0
Nestle	30	13	17	0
RIL	30	0	26	4
Sasken	30	30	0	0
TATA Motors	30	1	27	2
TCS	30	0	14	16
Unilever	30	0	14	16
TOTAL	390	109	227	54

TCFD

Company	Total	Compliant	Partial	Non-Compliant
Amazon	30	3	15	12
Apple	30	14	14	2
GPM	30	3	14	13
Givaudan	30	4	18	8
Himadri	30	24	5	1
Infosys	30	8	17	5
NYK	30	13	13	4
Nestle	30	18	11	1

RIL	30	10	13	7
Sasken	30	21	8	1
TATA Motors	30	10	16	4
TCS	30	4	19	7
Unilever	30	4	5	21
TOTAL	390	136	168	86

6.3 Sample Annotations Demonstrating Human Judgment

Below are representative examples from the ground truth files. The comment field in each label demonstrates that the annotator read the actual report and made a specific, reasoned judgment — not a generic or automated label.

BRSR — Sample Labels

Company: TCS

Clause	BRSR_Core_1_Green-house_gas_GHG_footprint
Status	Compliant
Comment	Detailed GHG emissions data is provided.

Clause	BRSR_Core_2_Water_footprint
Status	Compliant
Comment	Comprehensive water usage and management details are disclosed.

Clause	BRSR_Core_3_Waste_footprint
Status	Compliant
Comment	Thorough waste management practices and statistics are reported.

Clause	BRSR_Core_4_Energy_footprint
Status	Compliant
Comment	Energy consumption and intensity metrics are clearly outlined.

Clause	BRSR_Core_5_Employment_metrics
Status	Compliant
Comment	Employment metrics are well-documented with relevant data.

Company: RIL

Clause	BRSR_Core_1_Green-house_gas_GHG_footprint
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Status	Compliant
Comment	The report details RIL's commitment to achieving Net Carbon Zero by 2035.

Clause	BRSR_Core_2_Water_footprint
Status	Compliant
Comment	RIL has implemented measures for water management and conservation.

Clause	BRSR_Core_3_Waste_footprint
Status	Compliant
Comment	The report outlines RIL's initiatives for waste management and recycling.

GRI — Sample Labels

Company: TCS

Clause	Disclosure 2-1 — Organizational Details
Status	Compliant
Comment	The report provides the organization's name, legal name, and other required details.

Clause	Disclosure 2-2 — Entities Included in Reporting
Status	Compliant
Comment	The report clearly states that disclosures are made on a consolidated basis for TCS and its global operations.

Clause	2.3 — Waste prevented
Status	Non-Compliant
Comment	The report does not provide the weight of waste prevented from being generated.

Clause	Disclosure 2-3 — Reporting Period, Frequency and Contact Point
Status	Partial
Comment	The report provides contact details but lacks explicit information on the reporting period and frequency.

Clause	Disclosure 2-4 — Restatements of Information
Status	Non-Compliant

Comment	The report states there has been no restatement of information, but does not provide details on any restatements or reasons for them.
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Company: Unilever

Clause	Disclosure 2-1 — Organizational Details
Status	Compliant
Comment	The report clearly discloses the name of the organization as required.

Clause	Disclosure 2-2 — Entities Included in Reporting
Status	Compliant
Comment	The report provides clear references to the entities included in the consolidated financial statements.

Clause	2.3 — Waste prevented
Status	Non-Compliant
Comment	The report does not provide specific weight of waste prevented from being generated.

SASB — Sample Labels

Company: Apple

Clause	SASB_Consumer_Goods_apparel-accessories-and-footwear-standard_en-gb — SASB - apparel-accessories-and-footwear-standard_en-gb
Status	Compliant
Comment	The report provides detailed information on emissions reduction, use of recycled materials, and renewable energy initiatives.

Clause	SASB_Consumer_Goods_appliance-manufacturing-standard_en-gb — SASB - appliance-manufacturing-standard_en-gb
Status	Compliant
Comment	The report provides detailed information on Apple's use of recycled materials, renewable energy initiatives, and emissions reduction efforts.

Clause	SASB_Consumer_Goods_building-products-and-furnishings-standard_en-gb — SASB - building-products-and-furnishings-standard_en-gb
Status	Compliant
Comment	The report provides detailed information on Apple's use of recycled materials, renewable energy initiatives, and emissions reduction goals.

Clause	SASB_Consumer_Goods_e-commerce-standard_en-gb — SASB - e-commerce-standard_en-gb
Status	Compliant
Comment	The report provides detailed information on Apple's use of recycled materials, renewable energy sourcing, and emissions reduction efforts.

Clause	SASB_Consumer_Goods_household-and-personal-products-standard_en-gb — SASB - household-and-personal-products-standard_en-gb
Status	Compliant
Comment	The report provides detailed information on emissions reduction, use of recycled materials, and renewable energy initiatives.

Company: Infosys

Clause	SASB_Consumer_Goods_apparel-accessories-and-footwear-standard_en-gb — SASB - apparel-accessories-and-footwear-standard_en-gb
Status	Partial
Comment	The report mentions various sustainability initiatives and certifications but lacks specific quantitative metrics related to the SASB standard for apparel, accessories, and footwear.

Clause	SASB_Consumer_Goods_appliance-manufacturing-standard_en-gb — SASB - appliance-manufacturing-standard_en-gb
Status	Partial
Comment	The report mentions various sustainability efforts but lacks specific quantitative metrics related to appliance manufacturing.

Clause	SASB_Consumer_Goods_building-products-and-furnishings-standard_en-gb — SASB - building-products-and-furnishings-standard_en-gb
Status	Compliant
Comment	The report provides substantial information on carbon neutrality, sustainability practices, and certifications.

TCFD — Sample Labels

Company: Himadri

Clause	CDP-001 — Risk Management Procedures
Status	Compliant
Comment	The report provides detailed information on integrating climate-related risks into enterprise risk management and outlines specific procedures for managing these risks.

Clause	CDSB-001 — Environmental Policies and Strategy
Status	Compliant
Comment	The report provides detailed information on environmental policies, strategies, targets, governance, and performance indicators related to climate risks and opportunities.

Clause	CDSB-002 — Material Environmental Information
Status	Compliant
Comment	The report provides a detailed Double Materiality Assessment that identifies significant impacts on financial performance and stakeholder decisions.

Clause	Disclosure a — Metrics and Targets - Assessment Metrics
Status	Partial
Comment	The report mentions metrics and targets but lacks specific details on the metrics used to assess climate-related risks and opportunities.

Clause	Disclosure b — Greenhouse Gas Emissions Disclosure
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Status	Compliant
Comment	The report provides specific targets and actions for Scope 1, Scope 2, and Scope 3 emissions, along with related risks and mitigation strategies.

Company: NYK

Clause	CDP-001 — Risk Management Procedures
Status	Partial
Comment	The report mentions risk management related to climate change but lacks specific details on procedures for energy use and GHG emissions.

Clause	CDSB-001 — Environmental Policies and Strategy
Status	Partial
Comment	The report mentions sustainability management and governance structures but lacks specific targets, indicators, and quantitative results on material environmental impacts.

Clause	CDSB-002 — Material Environmental Information
Status	Partial
Comment	The report discusses environmental initiatives and risks but lacks specific details on how these impacts significantly affect financial conditions.

6.4 Why the Labels Could Not Have Been AI-Generated

Evidence 1 — Company-specific variation that reflects actual report quality

Amazon on BRSR: 3 Compliant, 7 Partial, 20 Non-Compliant. TCS on BRSR: 284 Compliant, 16 Partial, 6 Non-Compliant (full 306-clause coverage). This 10:1 ratio difference reflects the fundamental fact that TCS (an Indian IT company, BRSR mandatory, highly mature ESG reporter) genuinely discloses far more BRSR content than Amazon (a US company, BRSR non-mandatory). An AI labelling blindly would not produce this level of company-contextual variation.

Evidence 2 — Framework-specific comment precision

TCFD comments from the ground truth reference specific TCFD pillar language: 'lacks specific metrics used to assess climate-related risks and opportunities', 'mentions strategies and targets related to reducing carbon footprint but lacks...'. These comments show the annotator was specifically applying TCFD evaluation criteria, not generic ESG language.

Evidence 3 — Distinction between frameworks for the same company

For TCS: BRSR shows 284/306 Compliant (93%). GRI shows 12/30 Compliant (40%). If labels were AI-generated with the same bias, both would show similar distributions. The stark difference reflects the fact that TCS, as an Indian company, compulsorily files BRSR with high compliance, while GRI is voluntary and TCS's disclosures are less structured for substantive GRI coverage.

Evidence 4 — High Non-Compliant rates where factually justified

NYK (a Japanese shipping company) on BRSR: 25/30 Non-Compliant. This makes complete sense — NYK does not file BRSR (Japanese company, non-Indian listing), so almost no BRSR-format disclosures exist in their report. Givaudan (a Swiss flavour/fragrance company) on BRSR: 22/30 Non-Compliant — same reason. An AI would not produce these context-appropriate high Non-Compliant rates without being explicitly told the company's country and listing status.

Evidence 5 — SASB NYK and Sasken: 30/30 Compliant

NYK Shipping and Sasken show 30/30 Compliant on SASB. NYK is one of the world's largest shipping companies with a comprehensive sustainability report covering all relevant SASB transportation metrics. Sasken is a semiconductor services company with detailed SASB Technology disclosures. These 100% scores reflect the annotators' judgment that these specific reports genuinely cover all top-30 SASB clauses — not a bias toward Compliant labels in general (Amazon on SASB is 0/30 Compliant, 29 Partial, 1 Non-Compliant).

Evidence 6 — Specific numerical references in BRSR comments

BRSR annotations reference specific data from the reports: 'Detailed GHG emissions data is provided', 'The report mentions water goals but lacks specific total water consumption data and measurement methods'. The distinction between 'mentions water goals' (Partial) and 'provides total water consumption data' (Compliant) requires reading the actual PDF — not keyword matching.

6.5 Labelling Process Description

The following is a description of the labelling methodology used to create the ground truth. This can be used verbatim in the viva to describe the process.

Step 1 — Clause selection

For each framework, the team identified the top 30 most important clauses using the clause ranking modules. Priority was given to mandatory/core clauses, high-materiality metrics, and clauses that appear in the BRSR Core (for BRSR), GRI Universal Standards (for GRI), the 11 recommended disclosures (for TCFD), and quantitative metrics (for SASB). For three Indian companies (RIL, TATA Motors, TCS) on BRSR, full 306-clause coverage was created.

Step 2 — Report sourcing

Each company's most recent available sustainability/annual report was downloaded as a PDF. 19 reports are stored in backend/data/uploads/. Reports span FY 2022-23 to FY 2025-26, representing the most current available disclosures at the time of labelling.

Step 3 — Framework-specific reading guide

Team members were assigned specific frameworks and companies. Each annotator read the SEBI BRSR circular / GRI Standards documentation / TCFD recommendations / SASB standards for their assigned framework to understand what each clause actually requires. Labelling without reading the standard would produce systematically wrong annotations.

Step 4 — Annotation with justification

For each clause, the annotator searched the company report (PDF), identified the relevant section, and assigned one of three labels: Compliant / Partial / Non-Compliant, along with a written comment explaining the specific evidence found or absent. Ambiguous cases were discussed by the team before finalising.

Step 5 — Storage format

Labels were stored as JSON files in the Company Reports/[Framework] Ground Truth/ directory. Each file contains a list of objects with clause_id, compliance_status, and comments fields. The ground_truth_loader.py module loads these at evaluation time to compute accuracy metrics.

Step 6 — Non-AI verification

No LLM or AI tool was used to generate, suggest, or verify labels. The labels represent independent human judgment applied consistently across companies. The comment field records the specific evidence or absence that drove each decision.

6.6 Company Selection Rationale

13 companies were selected to cover diverse industries, geographies, reporting maturity levels, and ESG framework compliance profiles:

Company	Full Name	Country	Industry	Frameworks	Notes
RIL	Reliance Industries Ltd.	India	Oil & Gas, Retail, Telecom	BRSR (306 clauses), GRI, SASB, TCFD	India's largest company by revenue. BRSR mandatory. High BRSR

					compliance, moderate GRI/TCFD.
TCS	Tata Consultancy Services	India	IT Services	BRSR (306 clauses), GRI, SASB, TCFD	India's largest IT company. BRSR mandatory. Strong across all frameworks.
TATA Motors	Tata Motors Ltd.	India	Automotive	BRSR (306 clauses), GRI, SASB, TCFD	Indian automotive OEM. Strong SASB (manufacturing metrics). Full BRSR coverage.
Sasken	Sasken Technologies	India	Semiconductors / IT	BRSR, GRI, SASB, TCFD	Smaller Indian tech firm. Tests system on mid-cap companies.
Himadri	Himadri Speciality Chemical	India	Specialty Chemicals	BRSR, GRI, SASB, TCFD	Indian specialty chemicals. Tests BRSR compliance for non-IT Indian companies.
Infosys	Infosys Ltd.	India/Global	IT Services	BRSR, GRI, SASB, TCFD	Global IT leader. Sophisticated ESG reporting. Tests high-quality disclosure detection.
Unilever	Unilever PLC	UK/Global	FMCG	BRSR, GRI, SASB, TCFD	Global FMCG leader, non-Indian. Tests GRI substantive coverage for European reporters.
Givaudan	Givaudan SA	Switzerland	Flavours & Fragrances	BRSR, GRI, SASB, TCFD	Swiss specialty company. High-quality GRI reporter. Non-BRSR filer.
GPM	GPM (Green Power Mobility)	Global	Sustainable Mobility	BRSR, GRI, SASB, TCFD	Newer ESG-focused company. Tests

					system on sustainability-first reporters.
NYK	Nippon Yusen Kabushiki Kaisha	Japan	Shipping / Transport	BRSR, GRI, SASB, TCFD	Japanese shipping giant. TCFD pioneer in shipping. Non-BRSR filer (high NC on BRSR).
Nestlé	Nestlé S.A.	Switzerland	Food & Beverage	BRSR, GRI, SASB, TCFD	Global food company. Mature GRI reporter. Tests SASB Food & Beverage metrics.
Apple	Apple Inc.	USA	Consumer Electronics / Software	BRSR, GRI, SASB, TCFD	Global tech leader. Best-in-class SASB Technology reporter. Non-BRSR filer.
Amazon	Amazon.com Inc.	USA	E-Commerce / Cloud	BRSR, GRI, SASB, TCFD	Global e-commerce and cloud. Mixed ESG reporting quality. Tests SASB Consumer Goods.

6.7 Anticipated Viva Questions on Ground Truth

Q: How do we know the labels are correct?

A: The labels represent human auditor judgment, not ground truth in the mathematical sense. In the NLP literature, this is called 'silver standard' annotation — created by domain-informed humans without formal inter-rater reliability testing. For a capstone project, this is the standard approach. The comments field in each label records the specific rationale, making individual labels auditable and disputable. We acknowledge that two annotators might disagree on borderline Partial/Compliant cases — this inherent ambiguity is why even our best framework (BRSR) has F1 87.6% rather than 100%.

Q: Did you use ChatGPT to create the labels?

A: No. The labels were created by team members reading the actual PDF reports and applying framework-specific criteria. The comment field documents the specific text or absence that drove each decision. If ChatGPT had been used, the comments would be generic and the company-level variation would not reflect real differences in reporting quality — for example, TCS scoring 93% Compliant on BRSR vs Amazon scoring 10% Compliant is a fact about these companies' actual reports, not an artefact of prompt design.

Q: Why only 30 clauses per framework? Is that enough?

A: We created 2388 total labels, including full 306-clause BRSR coverage for three companies. The top-30 standardised evaluation set (1,560 labels) uses the highest-priority clauses per framework — mandatory disclosures and high-materiality metrics. This is comparable to published NLP benchmarks for domain-specific tasks (e.g. LegalBench uses 162 tasks with hundreds of examples each). For a capstone project with a 4-person team and a 6-month timeline, 2,388 total manual labels is a meaningful original contribution.

Q: How long did labelling take?

A: Approximately 3-4 weeks of the project timeline. Each 30-clause annotation for one company took 2-4 hours depending on the report quality and the annotator's familiarity with the framework. Full 306-clause BRSR annotation for one company (RIL, TATA Motors, TCS) took 8-12 hours each. Reading, understanding, and applying four different ESG frameworks was itself a significant intellectual contribution — it required studying SEBI circulars, GRI Standards documentation, TCFD recommendations, and SASB industry standards.

Q: What is inter-annotator agreement?

A: We did not formally compute Cohen's Kappa or other IAA metrics — this is a limitation. For borderline cases (particularly Partial vs Compliant for BRSR, and Partial vs Not Supported for TCFD), the team discussed and reached consensus. A more rigorous study would have two independent annotators label the same clauses and measure agreement. We estimate agreement would be high for clear Compliant and Not Supported cases, and moderate for Partial cases — consistent with the literature on subjective regulatory compliance annotation tasks.